Fundamentals of machine learning

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As you start working with AI, you might hear people use the terms artificial intelligence (AI) and machine learning interchangeably. While these concepts are closely related, there are significant differences between them.

So, what are those differences and why do they matter?

Understand artificial intelligence

When we say **artificial intelligence (AI)**, we're referring to a broad field focused on creating tools that can complete tasks usually associated with human intelligence. These tasks can range from writing emails to suggesting driving routes.

While AI may seem smart, it's important to recognize that its knowledge comes from the specific data it was trained on. This is where Machine Learning comes in.

Understand machine learning

One of the most common ways of training AI models is with machine learning, a technique that enables a tool to find patterns and learn from data on its own. It doesn't even need to be programmed for every possible scenario.

For example, let's say someone wants to use AI to sort zebras from horses. For this to work, they must first train a machine learning program so it can distinguish zebras from horses. They provide the machine learning program with thousands of images depicting both zebras and horses. As the machine learning program processes these images, it eventually learns to identify features of zebras, like stripes.

After being trained with machine learning, AI can then identify zebras from photos that weren't in its training set. This process is called **inference**.

Understanding the role machine learning plays helps you recognize that AI's performance is a direct result of the data it was trained on.

Approaches to machine learning

There are three common machine learning approaches used to develop AI:

1. **Supervised learning** is used to train AI from a large dataset that has been labeled by people. This technique is often used when there is a specific, known output in mind.
   - For example, a model is trained on millions of pictures that a person has explicitly labeled zebra or horse. AI learns the specific characteristics of each animal so it can automatically sort new images into those predefined categories.
2. **Unsupervised learning** is used to train AI from a dataset that has *not* been labeled by people. This technique is used to identify patterns and structures in data when there isn't a specific, known output in mind.
   - For example, a model analyzes a large, unlabeled dataset of photos containing horses and zebras. Without being told what a zebra is, the AI discovers patterns on its own, like striped or solid, and clusters images with similar features together.
3. **Reinforcement learning** is used to train AI through a trial-and-error process that is guided by a reward system. This technique is used to continuously refine how AI approaches a specific task or goal.
   - For example, a model is tasked with identifying zebras in a video. It has not received any training, so it has to make a guess. Each time the model guesses correctly, it receives a "positive reward." If the guess is wrong, it receives a "penalty." Over many attempts, AI learns to adjust its strategy to maximize its total reward, gradually mastering the task through experience.

Today, many AI models use a combination of all three machine learning approaches to create text, images, video, and more.

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